

SCOPE AND SOLUTION

Building a Loyalty Ecosystem worth growing into, together

By bringing PALO IT's delivery expertise and Open Loyalty's proven gamification engine, we open up possibilities for what NTUC Club's Loyalty Programme can become.

THE AMBITION | PHASE 3 & BEYOND

This is the end state we're building toward - not a feature list, but the moment the programme stops competing on discounts and starts competing on experience. Personalisation, play and community turn a points scheme into a lifestyle ecosystem members feel part of.

Personalised journeys

Offers and missions shaped by real member behaviour, not broadcast promos.

Play, not just earn

Streaks, challenges and badges turn everyday visits into habits worth repeating.

A living community

Referral loops and an experience marketplace extend loyalty beyond the transaction.

THE TOOLKIT: PROVEN LOYALTY AND GAMIFICATION MECHANICS THAT MAKE IT REAL



Points

Currency for every action



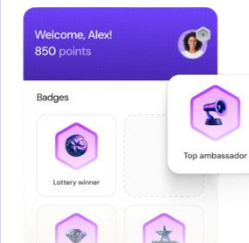
Tiers

Recognise your best members



Coupons

Targeted, time-boxed offers



Badges

Status beyond spend



Automations

Right message, right moment



Wheel of Fortune

Surprise & delight



Leaderboards

Friendly competition



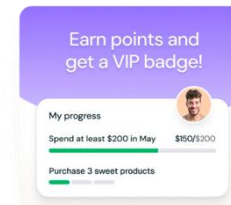
Referrals

Members as growth engine



Streaks & Challenges

Visits become habits



SCOPE AND SOLUTION

Tackling Phase 1 First

As a first step, PALO IT will build the Member and Merchant platforms as well as the integration layer bridging it to Open Loyalty's engine (API, webhooks, admin dashboard), creating a seamless launch experience for users.

This is simple by design to hit the 1 Dec launch date – a free, two-tier points programme before future scaling.



Member Enrolment & Profile

- SingPass/MyInfo as baseline sign-up, plus Google, Apple, email & mobile OTP
- Digital membership card (QR/barcode); self-service profile & PDPA consent controls



Free, Two-Tier Points Programme

- Free to join - Tier 1 Member & Tier 2 Premium, with NTUC union member auto-placement
- Admin-configurable qualification, retention & earn rates (e.g. \$1 = 1 pt baseline)



Tenant Transaction Capture

- QR scanner/device at each participating tenant
- Auto points crediting, with refund/reversal and exclusion-rule handling



Rewards & Redemption

- Points-to-voucher conversion, welcome bonus & birthday multipliers
- Monthly parking entitlement (physical coupon); 12-month expiry on points & rewards



Core Systems Integration

- Bi-directional Salesforce CRM sync; marketing automation platform integration
- Data warehouse extracts; CIAM/ SSO with SingPass authentication
- Data integration with the NTUC HQ system (union member validation & migration)



Promotions & Reporting

- Member-only, tier/segment-targeted promotions
- Self-service dashboard

• PHASE 1

Personas and Journeys

MEMBER

Clubhouse guests, union members, families

Enrolment · Singpass, email or social sign-in, CRM ID match, digital card

Earn · scan card at payment, upload receipt, tier-based accrual

Redemption · browse rewards, QR voucher issued and consumed in-club

MERCHANT / TENANT

Tenant staff and cashiers across the various NTUC Club touchpoints

Onboarding · scanner device issued, account activated, flow training

Scan and confirm · member QR scanned at checkout, spend confirmed

Fallback · offline queueing and manual entry so no sale is lost

PROGRAMME ADMIN

NTUC Club Loyalty, Marketing and CRM Owners

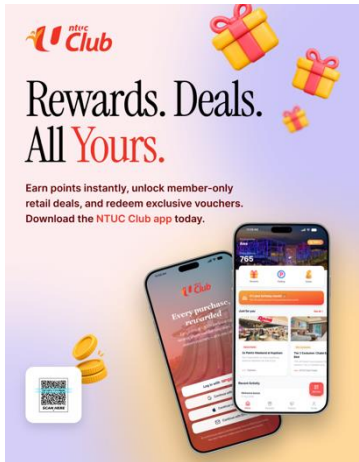
Configure · tiering, accrual and expiry rules, member segments

Campaigns · trigger events, promotions, personalised communications

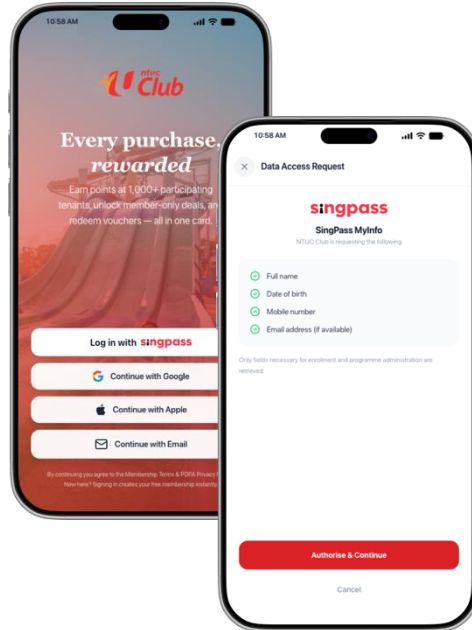
Monitor · dashboards for enrolments, redemptions and transactions

• PHASE 1 JOURNEYS

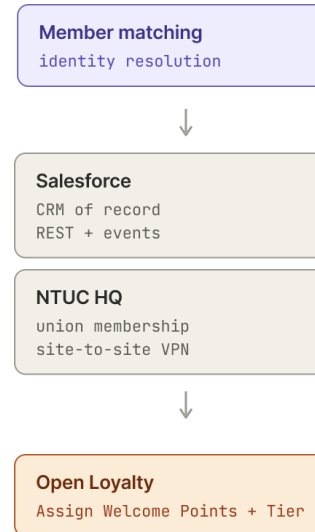
Member: Enrolment



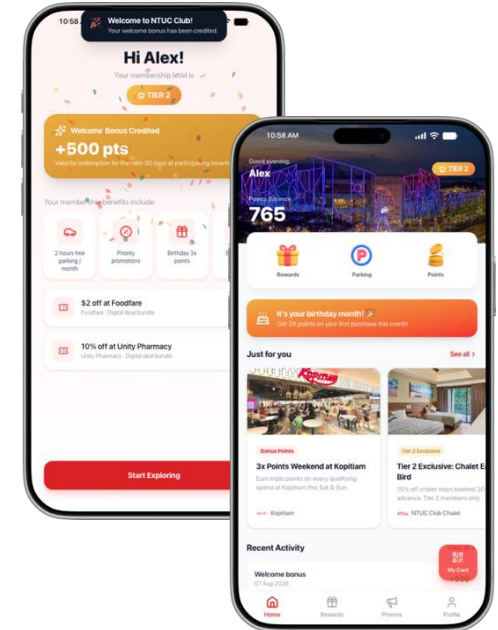
Spark the moment
Posters and QR prompts turn a casual visit/comms into an instant app download.



Sign In, Instantly
Singpass, Google, Apple ID or email - members join with a login they already trust.



One Member, One ID
Identity is resolved and matched across CRM and NTUC HQ System (Athena) with union members mapped automatically.



Welcomed like a VIP
Members land straight into their digital card with welcome points already credited. Union members auto-placed into Tier 2

• PHASE 1 JOURNEYS

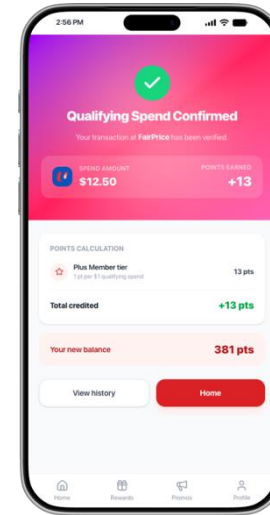
Member: Earn



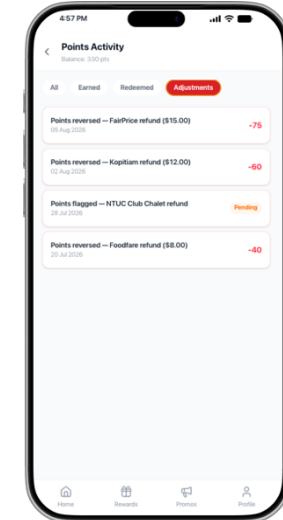
Show Up and Shop
Members browse and buy exactly as they do today.



Tap to Identify
One QR presented at checkout.
The digital Loyalty Card does the work.



Points, in Real Time
Earn rates apply instantly by tier and spend.



Reversal
Refunds are detected and the matching points reversed without any manual clean-up.

• PHASE 1 JOURNEYS

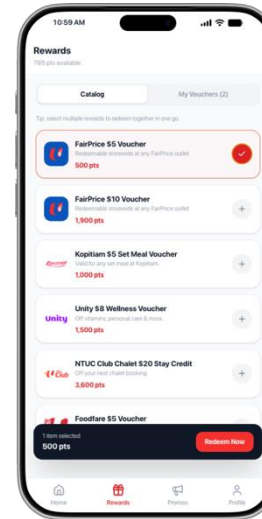
Member: Redemption



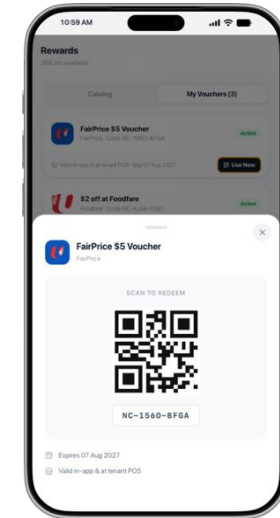
A member spots a reward they want and asks for it in the moment



One Scan Away
Staff scan the digital loyalty card.



Reward, Delivered
A digital voucher with its own QR is awarded instantly and auto-invalidated the moment it's used.



Instant Savings
The transaction is discounted on the spot based on the voucher. Value the member sees immediately.

- PHASE 1 JOURNEYS

Merchant / Tenant: Onboarding & Transactions

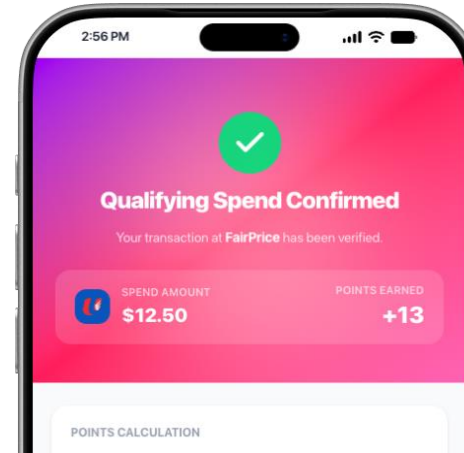


Staff Onboarding

Tenant staff issued scanner device/app; account activated and trained on the flow

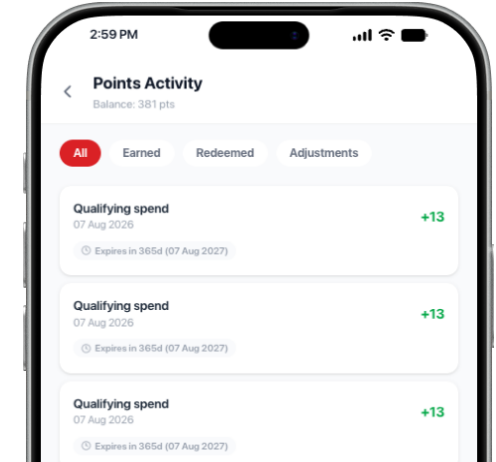
Scan Customer QR

Staff scans the member's digital QR at checkout with the provided scanner/device



Confirm Transaction

Qualifying spend confirmed; points calculated per active tier and campaign rules



Points Land Instantly

Credited to the member in real time, with confirmation shown to both staff and member.

Offline Fallback - if the scanner is offline, the transaction is queued locally and points are credited automatically once connectivity resumes; manual entry remains available as a backup so no sale is lost.

- PHASE 1 JOURNEYS

Programme Admin: Configure and

The image displays two screenshots of the Programme Admin interface. The left screenshot shows the 'Tier set information' and 'Define conditions' steps. The 'Define conditions' step includes a warning about resetting threshold values, a list of conditions (Total spending (SGD), Active units, Total earned units, Total spending (SGD), Months since joining the program, Cumulative earned units), and a table for 'Tier set 1' with columns for 'Active units' and 'Months since joining the program'. The right screenshot shows the 'Campaign trigger' and 'Rules & Effects' steps. The 'Campaign trigger' step includes fields for 'Campaign name (en)', 'Campaign description (en)', 'Campaign display order', and 'Start & end campaign date'. The 'Rules & Effects' step includes a 'Select automation campaign trigger' section with options like 'Daily', 'Weekly', 'Monthly', 'Member's birthday', and 'Membership anniversary', and a 'Basic settings' section with fields for 'Automation campaign name (en)', 'Automation campaign description (en)', 'Start date', and 'End date'.

Set Up Membership Tiers

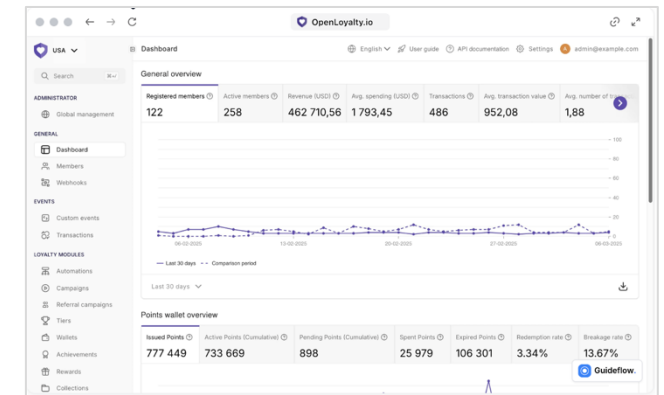
- Configure tier structure & thresholds
- Auto upgrade union members to Tier 2 with segmentation rules live in the admin console, not in code.
- Set qualification & downgrade rules

Configure Earn Rules

- Set base earn rate & exclusions
- Automate welcome & birthday bonuses

- PHASE 1 JOURNEYS

Programme Admin: Configure and Manage



Build Reward & Voucher Catalogue

- Create vouchers, discounts & coupons
- Set redemption limits & expiry

Manage Points Wallets

- Set points expiry & pending periods
- Auto-reverse points on refunds

Monitor Dashboard & Export Data

- Track members, spend & tier movement
- Export data feeds for CRM & warehouse

Technical Approach

Solution Architecture & Hosting

TECHNICAL APPROACH

Proposed Tech Stack

Front End & Back End



ReactJS

Progressive Web App



Flutter

Mobile App



Node.js

NestJS framework

Storage & Database



PostgreSQL

Primary data store



Redis

Caching layer



Azure Blob Storage

File storage

Cloud & DevOps



Terraform

Infrastructure as Code



Microsoft Azure

Cloud Platform



GitHub Actions

CI/CD pipelines



GitHub Enterprise

Source control

Authentication



singpass



Google Login



Apple Login

AI Assistance



GitHub CoPilot

AI Pair Programming

Testing



Playwright

End-to-end testing

Security



Microsoft Defender

Threat protection

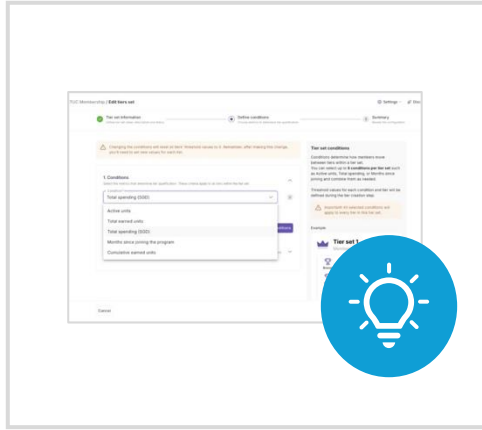


Microsoft Sentinel

SIEM & monitoring

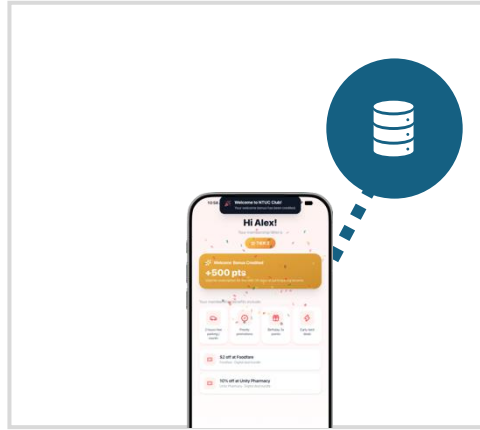
• TECHNICAL APPROACH

A dynamic loyalty programme that integrates with your systems



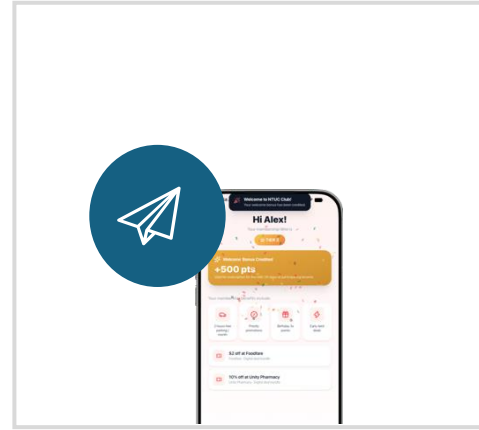
Your Customer Engagement Hub

Develop new ways to engage your customers directly in the loyalty portal without code. Every campaign can be simulated and tested in a separate environment



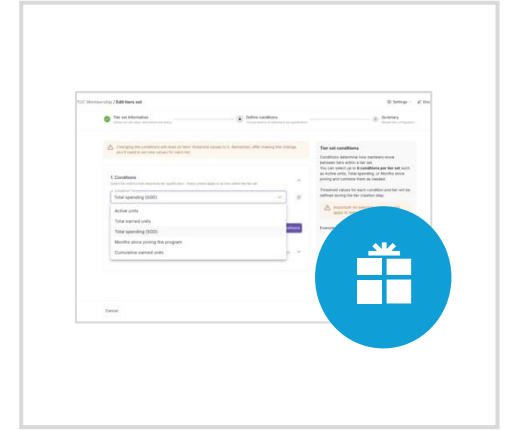
One Member, One Identity

Deterministic matching against Salesforce CRM and NTUC HQ at enrolment, resolving to link, create or exception queue; canonical ID and cross-reference layer keep the member addressable across app, loyalty platform, CRM and warehouse



Event-driven Communications

Outbound webhooks on every member and ledger mutation delivered as triggers to NTUC Club's marketing automation platform



Dynamic Rewards Configuration

A configurable catalogue: vouchers, tenant-funded offers, welcome and birthday rewards, seasonal drops. Add, retire or reprice rewards as the programme learns what members respond to

• TECHNICAL APPROACH Solution Overview

Member and Tenant App

Member app

PWA app for registration and display of digital loyalty card, member points and rewards catalog

Tenant app

PWA app for publishing transactions to the loyalty platform

Authentication

Members download the app and enrol or Sign in with Singpass / Google / Apple or Email

Backend Services

Profile Service

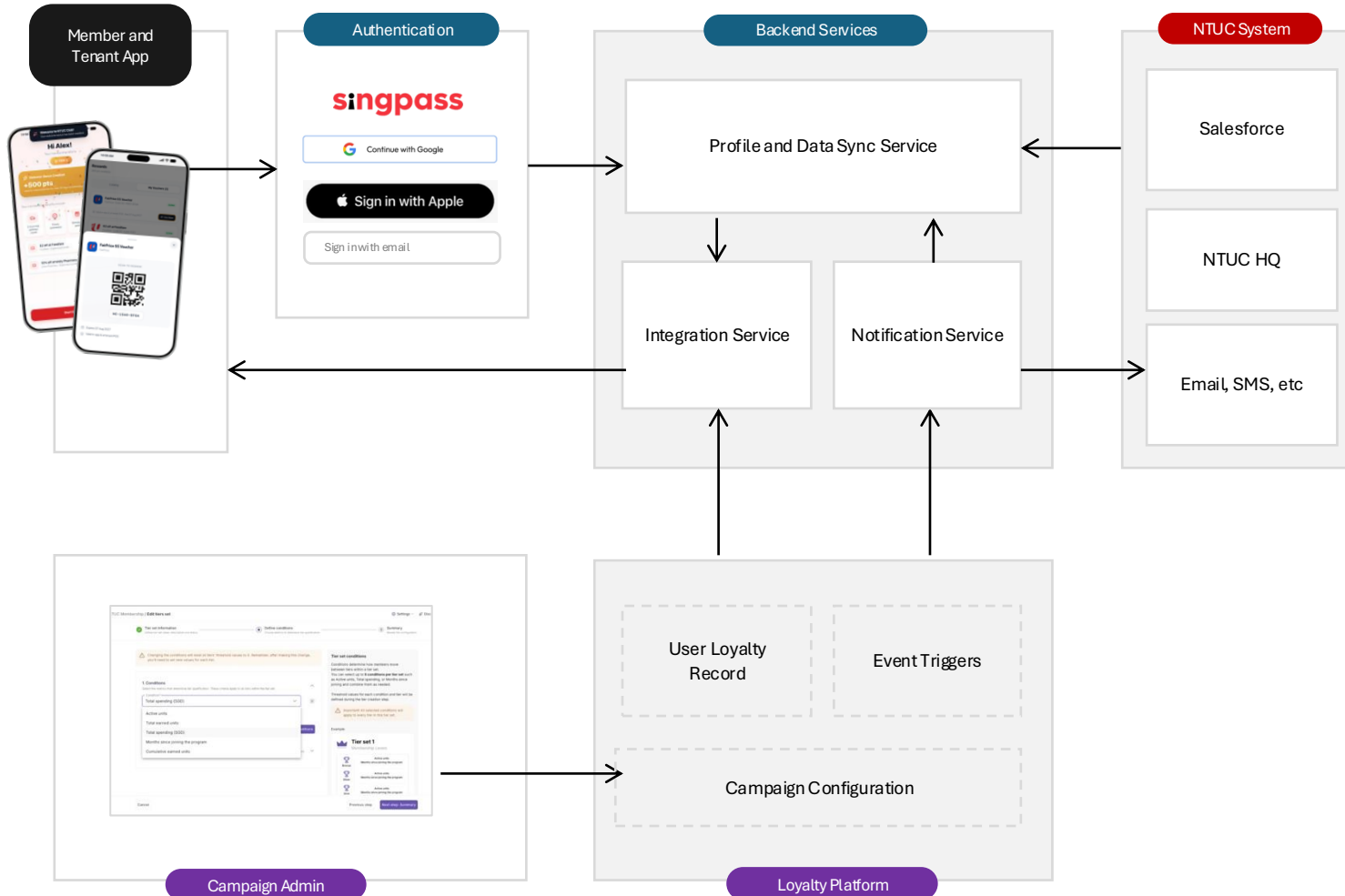
A service responsible for storing the users' loyalty profile, then mapping the loyalty id to user ids across multiple systems of record. Used by the notification service when dispatching messages to get the channel ID. Publishes loyalty data like total points and tier to multiple systems of record

Integration Service

Integrates the loyalty engine to the applications and publishes transactions from the tenant app into the loyaltyplatform

Notification Service

An API service that accepts webhook requests from OpenLoyalty and creates notifications that are sent to the marketing platform.



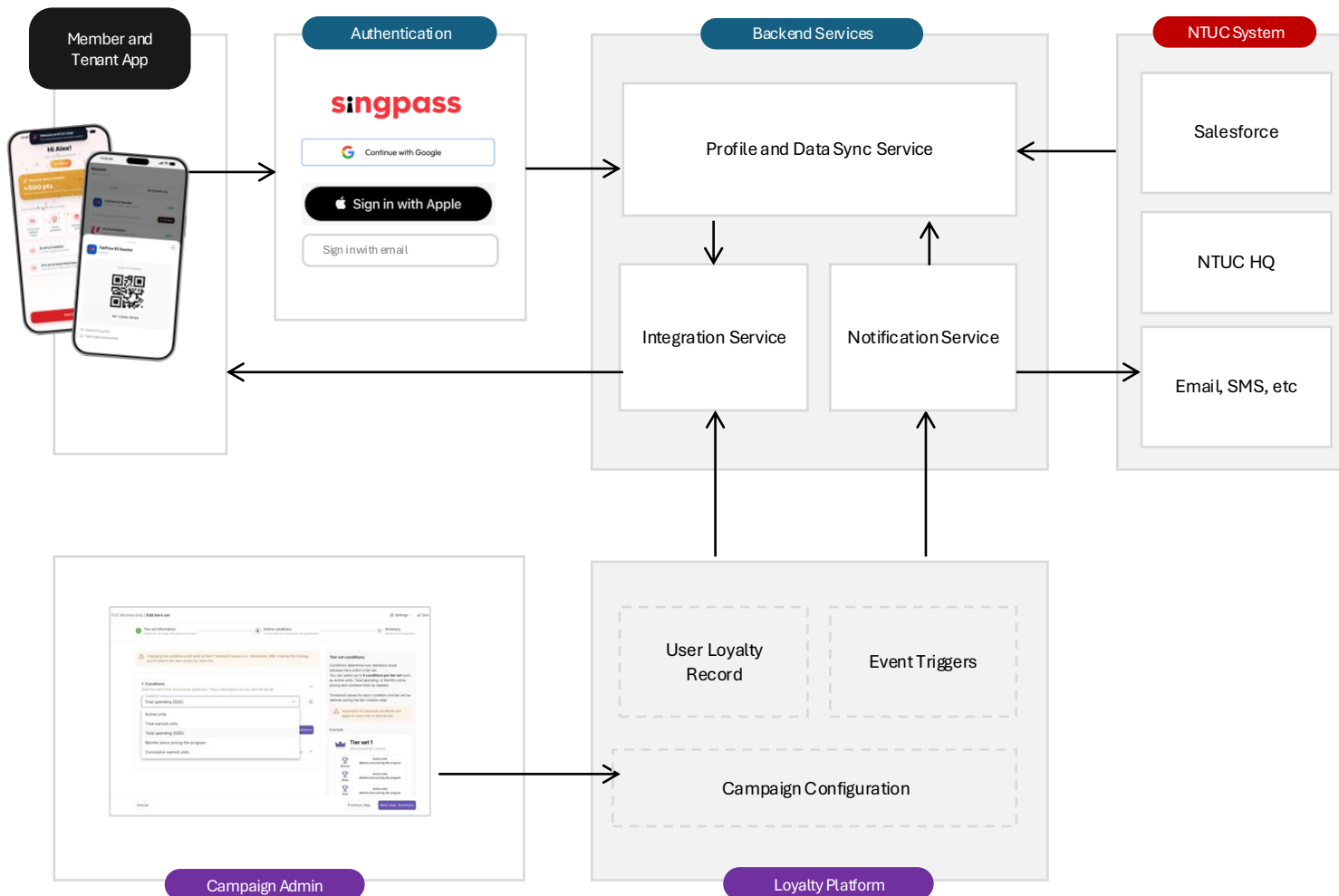
• TECHNICAL APPROACH Solution Overview

Loyalty Platform

Main engine that handles transaction ingestion, points assignment and redemption, rules configuration

Campaign Admin

A web app used by NTUC Club staff to configure the tiering system, configure the rewards as well as create and track campaigns .

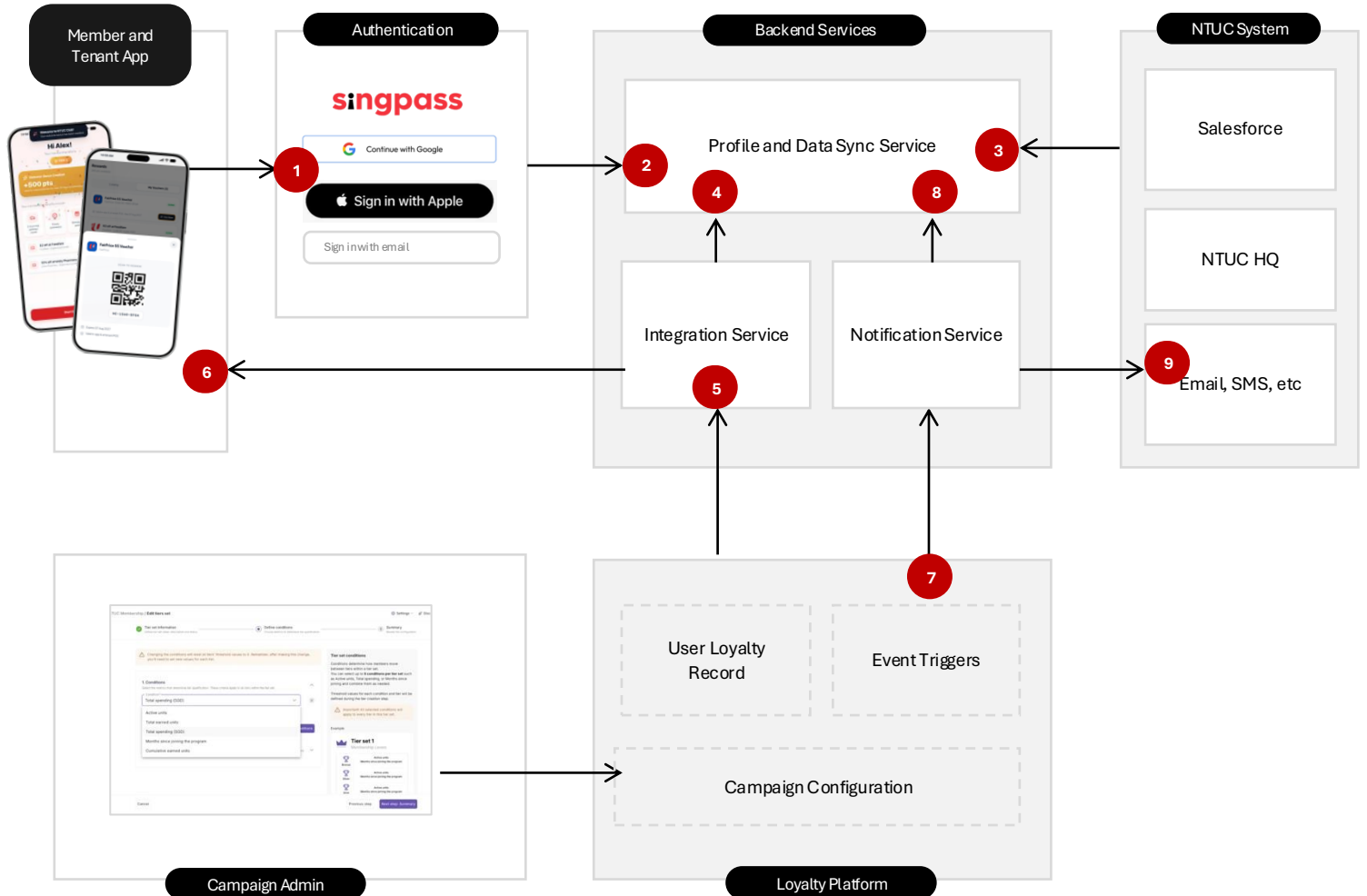


• TECHNICAL APPROACH

Member Registration

Member Registration

1. User authenticates through any of the available channels
2. User successfully authenticated. The profile service initiates profile matching.
3. The user is matched to a salesforce Id and an NTUC Id using the SingPass id or email and/or another field
4. A loyalty ID is assigned, the ID mapping is stored. The integration service is called to create the member record with NTUC membership status included.
5. The member is created in the loyalty platform, and the welcome points are awarded. Auto promotion happens if the user is an NTUC member
6. User lands on dashboard and sees welcome points
7. Enrolment successful event is dispatched
8. Notification services matches loyalty id to channel id
9. Message dispatched to marketing automation



• TECHNICAL APPROACH

Earn, Redeem, Reconcile

Earn / Redeem

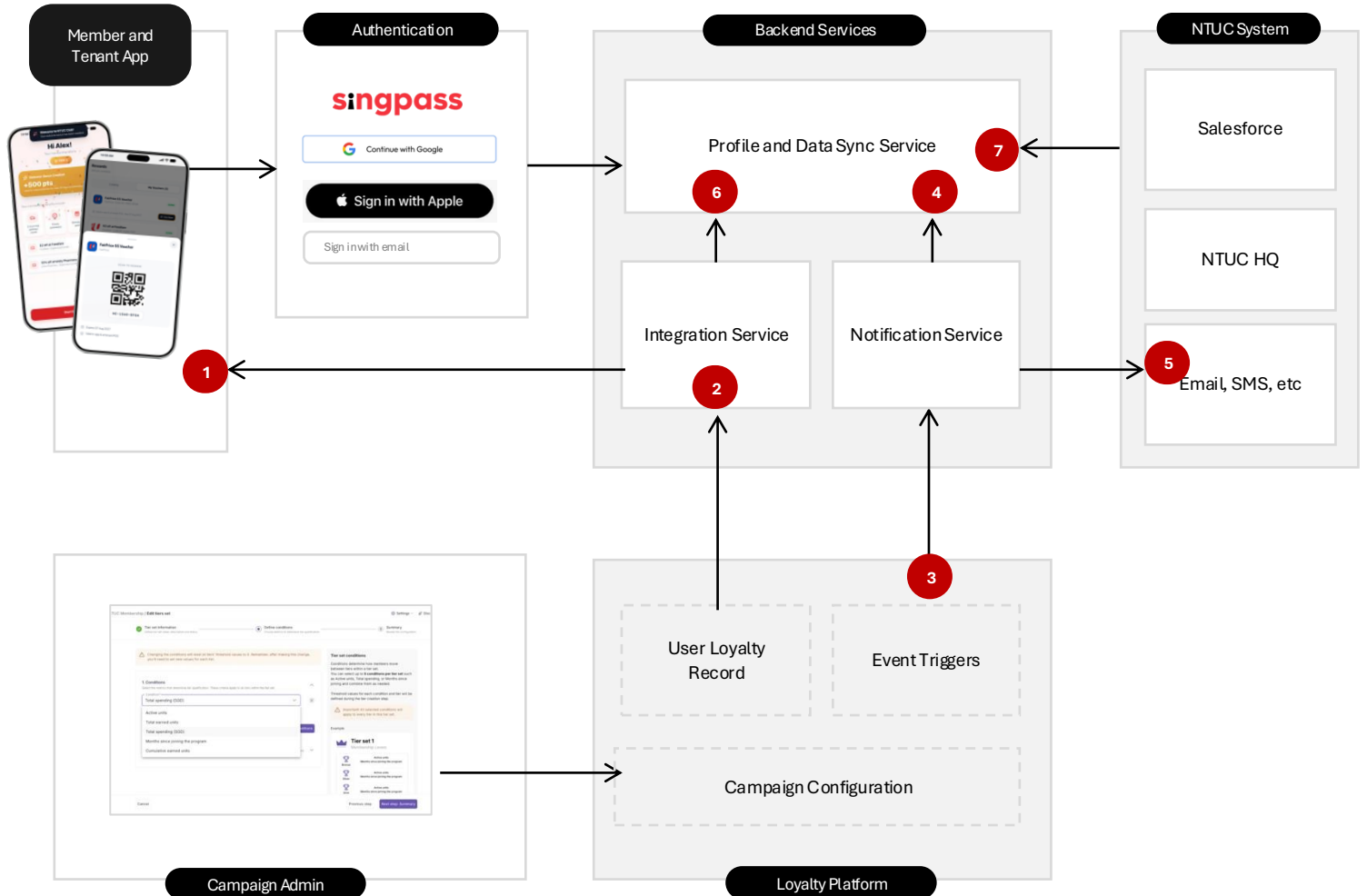
1. Customer completes a purchase, presents loyalty card. Tenant scans loyalty card. Tenant app calls integration service to post the transaction.
2. The transaction is picked up by the integration service and calls the loyalty platform. The loyalty platform calculates and checks from events such as tier promotion
3. Event triggers dispatched

Notify

4. Notification calls profile service to match loyalty id with channel id.
5. Notification service dispatches message to marketing automation

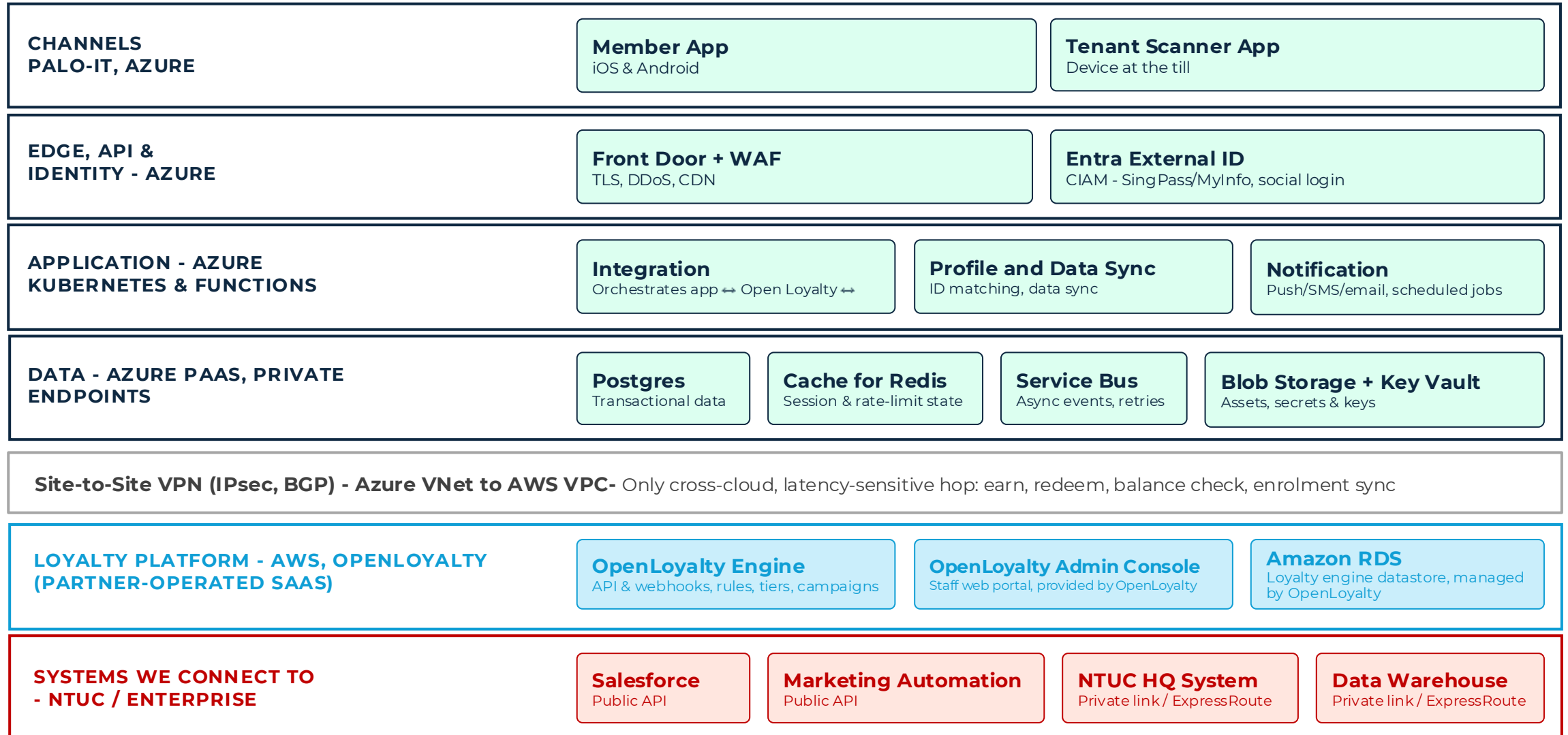
Reconciliation

6. Profile service fetches loyalty records
7. Profile service updates NTUC System with loyalty data



• TECHNICAL APPROACH

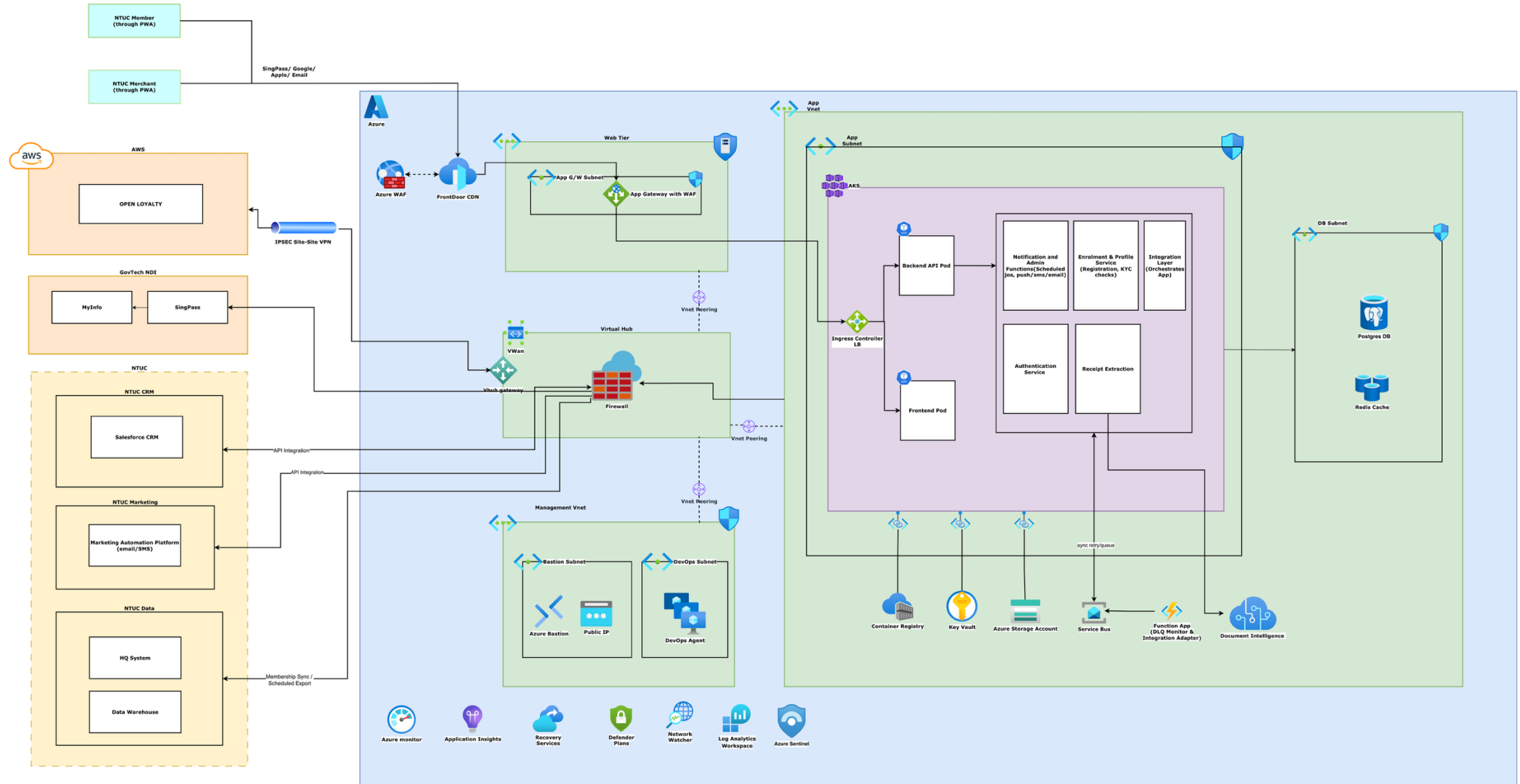
System Landscape Diagram



• TECHNICAL APPROACH

Application Network Architecture

For a higher-resolution image, please refer to the architecture diagram attached separately.



TECHNICAL APPROACH

Tenant Solution & Onboarding

- **Scan with any mobile device**
 - For phase 1, the tenants will be able to login to the tenant app through any device, scan the loyalty id, then input the transaction amount in order to complete a transaction.
- **Hardware Integration (Phase 2 – For exploration)**
 - Hardware solutions will be explored to record richer transaction data such as SKUs, transaction ID and other details coming from the POS
- **Training**
 - Digital training materials will be provided to each tenant in order to educate staff in performing earn, redemption, reversal and offline fallback
- **App Access**
 - Every new tenant will be provided access to the tenant app, where they can administer earn and redeem transactions

